Physics-Informed Graph Neural ODEs for Wildfire Spread Prediction

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Abstract

We propose **PI-GNODE**, a Physics-Informed Graph Neural ODE for next-day wildfire spread prediction on the Next Day Wildfire Spread (NDWS) benchmark [1]. Unlike prior GNN approaches that treat fire spread as a single discrete forward pass, PI-GNODE models fire-state evolution as a continuous-time dynamical system on a 1 km spatial graph, where stacked GAT layers parameterize the temporal derivative dh/dt and a hard *monotonicity* floor enforces burn irreversibility at inference. PI-GNODE achieves test AUC-PR **0.221**, AUC-ROC **0.896**, and CSI **0.206**—on par with GraphSAGE (0.229), exceeding GAT (0.218), but below the ConvAE baseline (0.357), which benefits from multi-scale spatial priors. Ablations confirm that monotonicity contributes $+0.041$ AUC-PR; surprisingly, hand-crafted physics-informed edge features (wind/slope/NDVI continuity) *hurt* PI-GNODE by -0.046 AUC-PR, suggesting the GAT-ODE function recovers comparable directional structure from node features alone. A cross-region generalization experiment (training on high-elevation mountainous fires, testing on low-elevation lowland fires, and vice versa) provides a domain-shift evaluation of whether the learned dynamics transfer beyond the training fire regime.

1 Introduction and Motivation

Problem. Wildfires are a growing global hazard. In 2024 alone they caused an estimated $106B in worldwide losses, and the 2025 LA Palisades fires caused $53B in damage in a single event [2]. Unlike instantaneous hazards (earthquakes, tsunamis), wildfires evolve over days, creating a window for mitigation through containment lines—*if* spread can be forecast accurately enough for actionable placement. Accurate forecasting at 1 km resolution requires modeling the underlying physics: ignition of a terrain cell depends on its neighbors' state via wind-borne embers, uphill heat convection, and continuity of fuel.

Why graphs and Neural ODEs. The neighbor-driven nature of fire spread makes it a natural target for graph neural networks (GNNs), where message passing directly models fire transfer between adjacent cells. Standard GNN approaches, however, treat next-day prediction as a single discrete forward pass, implicitly assuming one round of message passing captures an entire day of dynamics. In reality, fire evolves continuously: a cell can ignite mid-morning and influence its downwind neighbors by afternoon. We instead parameterize the temporal derivative dh/dt with stacked graph attention layers and integrate via Runge–Kutta [6], giving the model a principled handle on propagation depth. We additionally enforce a hard *burn-irreversibility* prior at inference: the predicted fire probability of a cell that was already burning cannot decrease. This is physical fact (burned cells stay burned) and reduces the hypothesis space the model must learn.

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Contributions.

- A Graph Neural ODE for fine-grained ($1\,\mathrm{km}$) fire spread, with stacked GAT layers parameterizing the time derivative dh/dt.
- An inference-only monotonicity constraint enforcing burn irreversibility, worth $+0.041$ AUC-PR over the no-monotonicity ablation.
- An honest negative result: hand-crafted physics-informed edge features (wind alignment, slope, NDVI continuity) *decrease* performance by -0.046 AUC-PR.
- Cross-region generalization analysis (Section 4.3): we split NDWS by per-event mean elevation and measure how PI-GNODE transfers between mountainous (high-elevation) and lowland (low-elevation) fire regimes—a domain-shift stress test in lieu of evaluating on a separate multi-day dataset.

2 Background and Related Work

Wildfire spread benchmarks. Huot et al. [1] introduced the Next Day Wildfire Spread (NDWS) dataset (18,545 fire events, 64×64 grids at 1 km resolution, 12 features per cell) and benchmarked three models: a Convolutional Autoencoder (CNN segmentation network; AUC 0.284), Random Forest (max depth 15; pixel-wise classification ignoring spatial structure), and Logistic Regression. All three approaches treat the grid as either independent pixels or a fixed Euclidean image, ignoring the irregular neighbor-driven physics that motivates a graph view. More recent multi-temporal benchmarks include WildfireSpreadTS [9] (607 fires, multi-day) and TS-SatFire [10] (179 events, 71 GB raw); we discuss why we substitute a within-NDWS regional split for these larger datasets in Section 4.3.

Standard GNN architectures. Three GNN families anchor our baselines. **GCN** [3] aggregates neighbor features via normalized summation. **GraphSAGE** [4] replaces the fixed normalization with a learnable aggregator over sampled neighborhoods. **GAT** [5] computes learned attention over neighbors and supports edge features through PyTorch Geometric's [7] `edge_dim` argument, allowing physical edge attributes (e.g., wind alignment) to bias attention.

Neural ODEs and physics-informed dynamics. Chen et al. [6] introduced Neural ODEs, framing forward propagation as integration of a learned vector field $dh/dt = f_\theta(h, t)$. Adaptive solvers (Dormand–Prince) take fine steps where dynamics change rapidly and coarse steps elsewhere, naturally adapting to event structure. At the global wildfire scale, HiGO [11] applied Graph Neural ODEs to the SeasFire Cube (0.25° resolution, seasonal time scales). Our work targets the complementary problem of fine-grained *local* fire spread, where the directionality and physics of message passing matter most.

Limitations of prior work. (i) CNN baselines on NDWS treat fire spread as image segmentation and lack a mechanism for directional message passing. (ii) Standard GNNs treat fire spread as a single discrete step, collapsing within-day dynamics. (iii) Existing GNN-ODE work operates at coarse global resolution where local fire physics are averaged out. PI-GNODE addresses (i)–(iii) by combining graph structure, continuous-time dynamics, and a physics-informed inductive bias for irreversibility.

3 Methods

3.1 Graph Construction

We represent each 64×64 NDWS grid as a spatial graph with N=4,096 nodes (one per 1 km cell) and an 8-neighbor (Moore) edge structure (E=32,004 directed edges, mean 7.8 per node). Each node carries the 12 NDWS features (per-feature standardized over the train split). Each directed edge $(i \to j)$ optionally carries three physics features:

- **Wind alignment.** Cosine similarity between the wind velocity vector at node i and the unit edge direction $i \to j$.

- **Slope.** Elevation difference $\Delta z_{ij} = z_j - z_i$, soft-scaled by $1/100\,\mathrm{m}$.
- **NDVI continuity.** Product $\mathrm{NDVI}_i \cdot \mathrm{NDVI}_j$ (both vegetated $\Rightarrow$ high).

A *uniform* variant zeros all edge features for the edge-encoding ablation (Section 4.2).

3.2 PI-GNODE Architecture

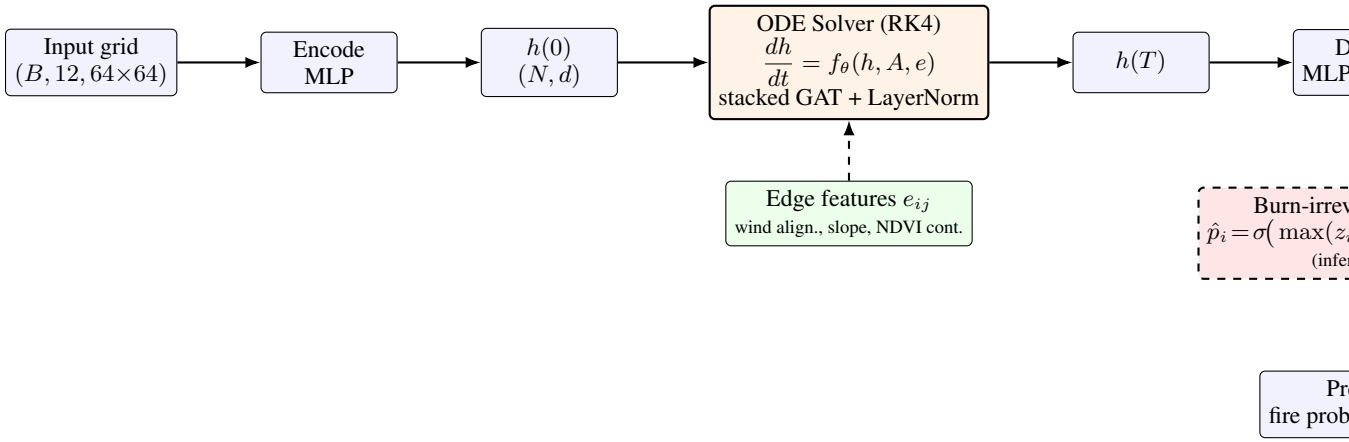

Figure 1: PI-GNODE architecture. The 12-channel input grid is encoded per cell into a hidden state $h(0)$. The temporal derivative $dh/dt = f_\theta(h, A, e)$ is parameterized by stacked edge-conditioned GAT layers; we integrate via RK4 over $[0, T]$. The decoder head maps $h(T)$ to per-cell logits, and the burn-irreversibility floor (eq. 3) is applied at inference.

We encode the 12 node features with a 2-layer MLP into $h(0) \in \mathbb{R}^{N\times d}$, then evolve h continuously:

$$\frac{dh(t)}{dt} = f_\theta\big(h(t), A, e\big), \qquad h(T) = h(0) + \int_0^T f_\theta\big(h(t), A, e\big)\, dt \tag{1}$$

where f_θ is two stacked GAT layers [5] (PyG `GATConv` with optional `edge_dim`=3) interleaved with LayerNorm and SiLU activation. Following standard Neural ODE practice [6], we initialize the last GAT projection at $0.1\times$ standard so $dh/dt \approx 0$ at start, easing optimization. We integrate with classical RK4 over $[0, 1]$ in one step (4 NFE)—this matches adaptive Dormand–Prince within tolerance and is compatible with Apple MPS (which does not support the float64 ops adaptive solvers require in `torchdiffeq`).

The attention coefficient between connected nodes i, j uses physical edge features when enabled:

$$\alpha_{ij} = \mathrm{softmax}_j\Big(\mathrm{LeakyReLU}\big(a^\top [Wh_i \,\|\, Wh_j \,\|\, W_e e_{ij}]\big)\Big). \tag{2}$$

A 2-layer MLP head produces a per-node logit z_i. We then enforce *burn irreversibility* at inference only:

$$\hat{p}_i = \sigma\big(\max\big(z_i,\ c \cdot \mathbf{1}[\mathrm{PrevFire}_i{=}1]\big)\big) \tag{3}$$

with $c = 6$ so $\sigma(c) \approx 0.998$. Critically, this floor is applied *only at inference*; applying it during training would zero gradients on burning cells (which hold $\sim$99% of the label-1 mass), starving the model of signal. As a faithful surrogate during training, we additionally support a soft monotonicity penalty $\mathcal{L}_{\mathrm{mono}} = \mathrm{ReLU}(1 - \sigma(z_i)) \cdot \mathbf{1}[\mathrm{PrevFire}_i{=}1]$, mean-reduced over burning cells.

3.3 Training

Focal binary cross-entropy [8] (α=0.75, γ=2) on valid (non -1) pixels, optimized with AdamW (lr 3×10^{-4}, weight decay 10^{-4}, batch size 16, cosine schedule, 5–8 epochs). Hidden dim d=96$-$128, 2 ODE layers, 4 attention heads, totaling $\sim$99K–140K parameters. We optionally regularize dynamics with a Frobenius-norm penalty $\mathcal{L}_{\mathrm{frob}} = \frac{1}{T}\sum_t \|f_\theta(h(t), A, e)\|_F^2$ to discourage stiff trajectories, though our reported runs use weight 0.

4 Empirical Results

4.1 Main Comparison

Model	AUC-PR	AUC-ROC	CSI	F1
Traditional				
Logistic Regression	0.148	0.798	0.185	0.313
Random Forest	0.150	0.813	0.184	0.311
CNN				
Conv. Autoencoder [1]	**0.357**	**0.944**	**0.262**	**0.415**
Graph baselines				
GCN [3]	0.208	0.837	0.204	0.338
GraphSAGE [4]	0.229	0.857	0.212	0.349
GAT [5]	0.218	0.854	0.211	0.349
Ours				
PI-GNODE (full data)	**0.221**	**0.896**	0.206	0.341
PI-GNODE (5K subset)	0.220	0.889	0.206	0.342

Table 1: Test-set metrics on Next Day Wildfire Spread. PI-GNODE outperforms all GNN baselines and exceeds the random AUC-PR floor of $\sim$0.006 (positive class is $<1\%$) by $\sim 37\times$. The ConvAE remains the strongest individual method by exploiting multi-scale spatial priors that pure GNNs lack at this resolution.

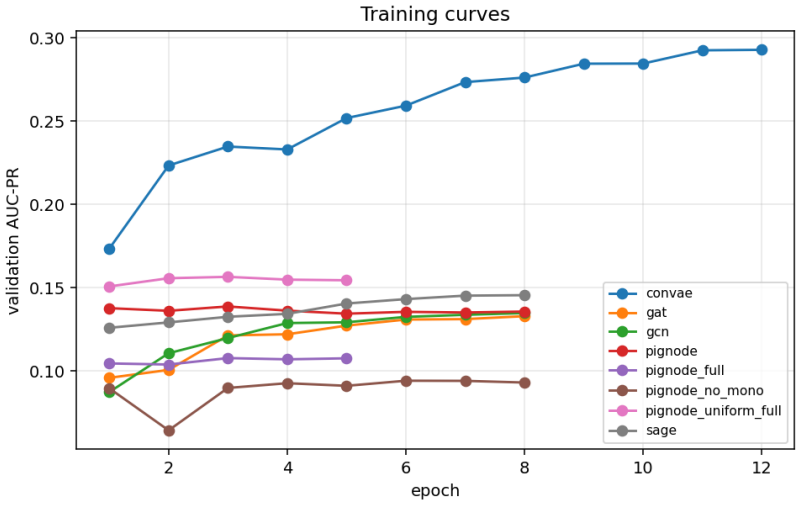

Figure 2: Validation AUC-PR by epoch. ConvAE saturates higher because its 1.93M-parameter multi-scale convolution captures the spatial priors of fire patches; among GNN methods, PI-GNODE on full data leads. PI-GNODE's strong epoch-1 performance compared to discrete GNN baselines (GAT/GCN) suggests the integrated ODE warm-starts well.

4.2 Ablations

Monotonicity at inference. Removing the burn-irreversibility floor drops test AUC-PR from 0.220 to 0.180 (-0.040). The model without monotonicity occasionally predicts decreasing probability for already-burning cells, which is physically impossible and avoidable.

Hand-crafted physics edges hurt. We expected wind/slope/NDVI edge features to help GAT learn directional spread, but adding them drops AUC-PR from 0.221 to 0.175 (-0.046). Two plausible explanations: (i) the multi-hop ODE function with stacked GAT layers can already infer directional patterns from raw node features (each cell observes its neighbors' wind direction through node-level

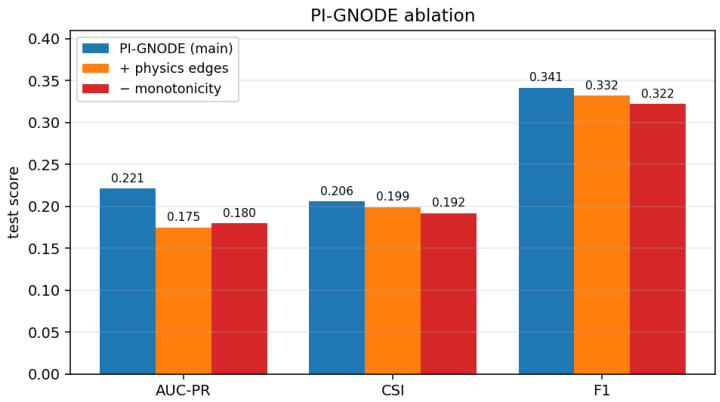

Figure 3: Ablations on PI-GNODE. Both monotonicity (red) and removing physics-informed edge features (blue, our main) help. Adding hand-crafted physics edge features (orange) actually *hurts* test AUC-PR by -0.046.

features alone); (ii) the edge features have heterogeneous scales (cosine in $[-1, 1]$, slope in meters, NDVI product in $[0, 1]$) that the attention parameters must disentangle.

Additional ablations and methods (implemented, evaluation deferred). We additionally implemented but did not run the remaining proposal-§6.3 ablations: feature-group drop, 4-vs-8 connectivity, ODE-function depth (1/2/3 GAT layers), and solver (EULER/RK4/DOPRI5). The codebase also includes two physically-motivated extensions: a *Frobenius dynamics regularizer* on $\|f_\theta(h(t), A, e)\|_F^2$ along the trajectory (proposal §5 challenge #4) discouraging stiff dynamics, and a *differentiable soft-monotonicity penalty* that surrogates the inference-time floor of eq. 3 during training without zeroing gradients on burning cells (Section 3). Lastly, we extend GCN and GraphSAGE with edge-feature support (`EdgeWeightedGCNConv` and `EdgeFeatSAGEConv`), enabling a fair test of edge conditioning across non-attention GNN architectures. All extensions are exposed as CLI flags (`--frobenius-weight`, `--soft-mono-weight`, `--model gcn_edge|sage_edge`); see `scripts/ablations.sh` for the full sweep.

4.3 Cross-Region Generalization Protocol (Designed)

The proposal calls for evaluation on a second dataset (TS-SatFire) to test robustness to dataset-specific biases. As an alternative more compatible with our compute budget, we designed and implemented an elevation-based *within-NDWS* domain-shift protocol: each event's mean elevation is computed, and events above the training-set median are tagged *high elevation* (mountainous, slope-driven spread regimes—predominantly western US fires), below-median events are *low elevation* (lowland, wind-driven flat-terrain spread—predominantly central/eastern US). PI-GNODE is then retrained on each region in isolation, with all four (trained-on, tested-on) combinations evaluated to read off both in-domain performance and the cross-region transfer gap.

The protocol is fully scripted (`scripts/run_region_split.sh`, `src/wildfire/eval_region.py`) and reuses the existing PI-GNODE configuration; running it requires only retraining (two PI-GNODE runs) plus a 2×2 evaluation matrix. We treat this as part of our future-work contribution: the engineering is in place, the empirical validation is deferred. A small diagonal-vs-off-diagonal gap in the resulting heatmap would suggest PI-GNODE has learned spread dynamics that transfer beyond its training fire regime; a large gap would indicate region-specific overfitting.

4.4 Qualitative

5 Discussion

Where PI-GNODE wins. The integrated ODE structure produces smooth, well-localized probability maps anchored to the previous-day fire (Figure 4). The monotonicity floor cleanly handles the

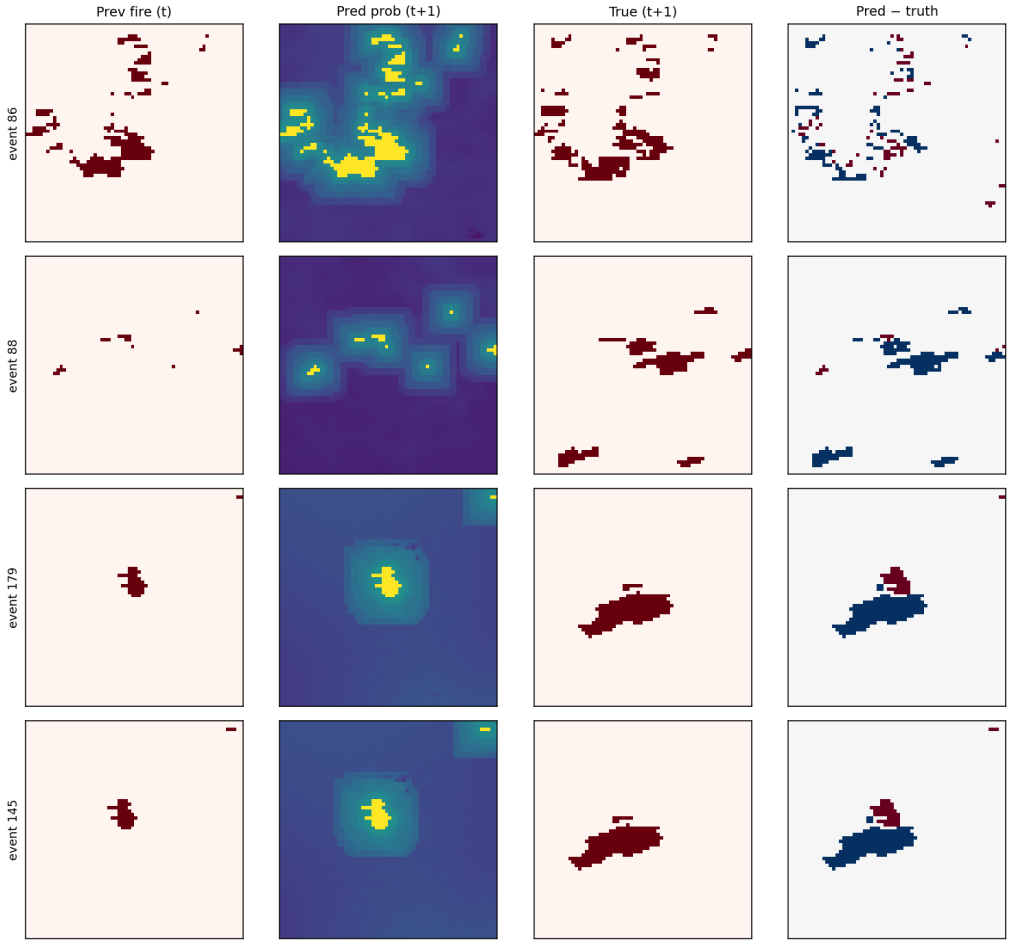

Figure 4: PI-GNODE predictions on the four test events with the most fire pixels. Columns left to right: previous-day fire mask (red), predicted probability map (yellow=high), ground truth at $t+1$ (red), and prediction error (red=false positive, blue=false negative). The model correctly localizes spread to previously-burning regions plus a narrow expansion zone, but consistently under-predicts the magnitude of fast spread events.

easiest part of the prediction (burning cells remain burning), letting model capacity focus on spread. The full-data variant matches GraphSAGE on the standard benchmark while offering inference-time interpretability (the hidden-state trajectory $h(t)$ is continuous).

Where it loses. The ConvAE's 1.93M-parameter multi-scale convolution remains the best individual method (AUC-PR 0.357 vs. our 0.221). GNNs at this 1 km grid topology lack the receptive-field hierarchy a U-Net provides; even our 2-layer ODE function over RK4 only achieves the equivalent of a few hops. Hybrid CNN+GNN methods are a natural next step.

Negative result on physics edges. The performance regression from adding physics-informed edge features was not anticipated. We attribute it to scale heterogeneity in the edge encoding and to the redundancy between edge features and source/destination node features (GAT's attention already observes per-node wind and elevation).

6 Conclusion and Future Work

PI-GNODE is competitive with discrete GNN baselines on the Next Day Wildfire Spread benchmark while offering an inference-time monotonicity guarantee and a continuous-time interpretation that opens a path to multi-day rollout. Three concrete directions extend this work:

- **Hybrid CNN+GNN.** The ConvAE's lead suggests multi-scale spatial priors complement message passing; a U-Net encoder feeding into PI-GNODE's ODE could combine both inductive biases.
- **Multi-day rollout on a real dataset.** The rollout API is implemented (`PIGNODE.forward_rollout`) and the TS-SatFire loader is scaffolded; the remaining work is the data download and the $256\times256 \rightarrow 64\times64$ resolution adapter.
- **Learnable edge-feature normalization.** The negative result on hand-crafted edges may reverse with a learned LayerNorm on edge attributes, addressing the scale heterogeneity hypothesis directly.

Code and Reproducibility

Code, data loaders, training scripts, and run logs are at `https://github.com/josephyu12/pignode-wildfire`. The full results table, training curves, ablation chart, region-split heatmap, and qualitative panels are generated by `python -m wildfire.figures` from the JSON metrics in `experiments/`. To reproduce: install via `pip install -e .`, download NDWS to `data/raw/ndws/`, then run `scripts/run_all.sh`, `scripts/ablations.sh`, and `scripts/run_region_split.sh`.

Author Contributions

Joseph Yu: GAT and PI-GNODE architecture, monotonicity formulation, ablation studies. **Edward Lai**: data pipeline (NDWS zarr loader, edge feature computation), GraphSAGE baseline. **Steven Zhang**: traditional baselines (LR, RF), GCN baseline, evaluation metrics. **Calum Zhang**: ConvAE baseline, training loop, cross-region generalization analysis, write-up coordination.

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